

Supplementary Information

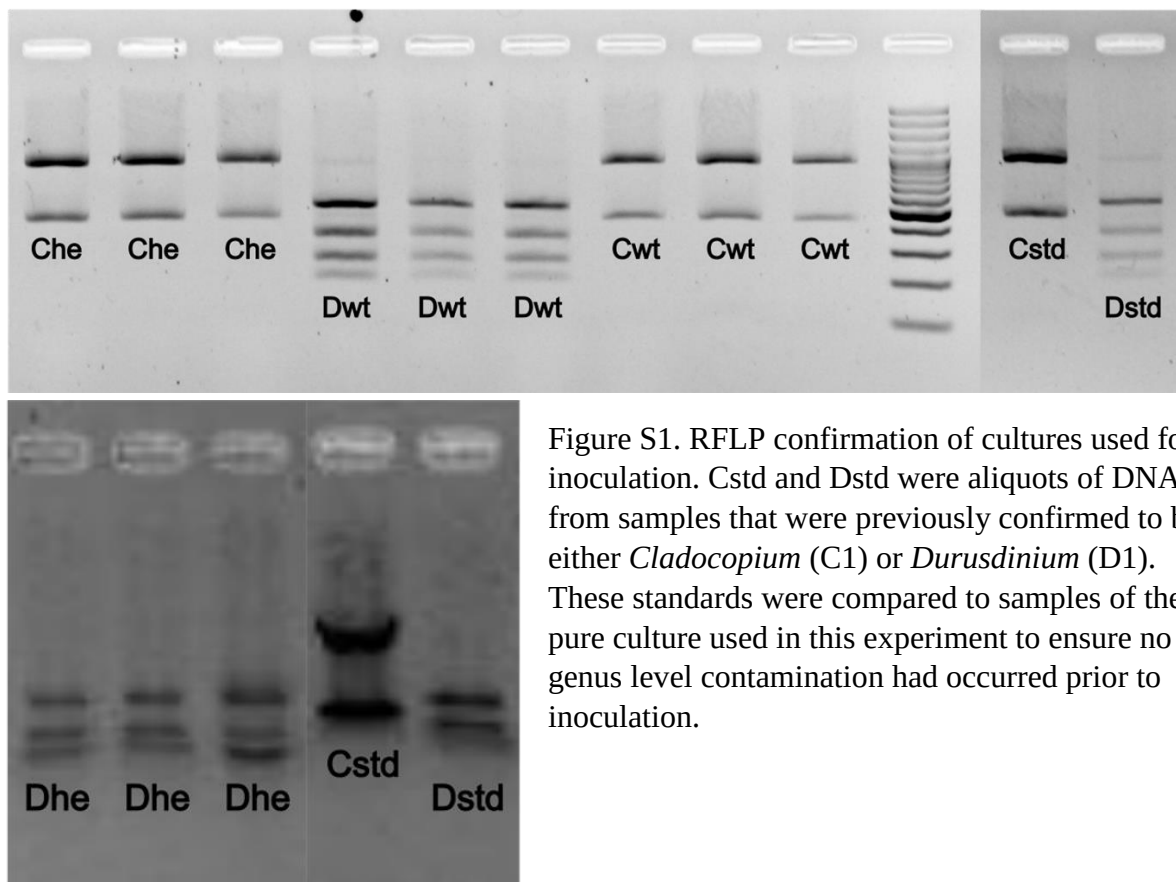


Figure S1. RFLP confirmation of cultures used for inoculation. Cstd and Dstd were aliquots of DNA from samples that were previously confirmed to be either *Cladocopium* (C1) or *Durusdinium* (D1). These standards were compared to samples of the pure culture used in this experiment to ensure no genus level contamination had occurred prior to inoculation.

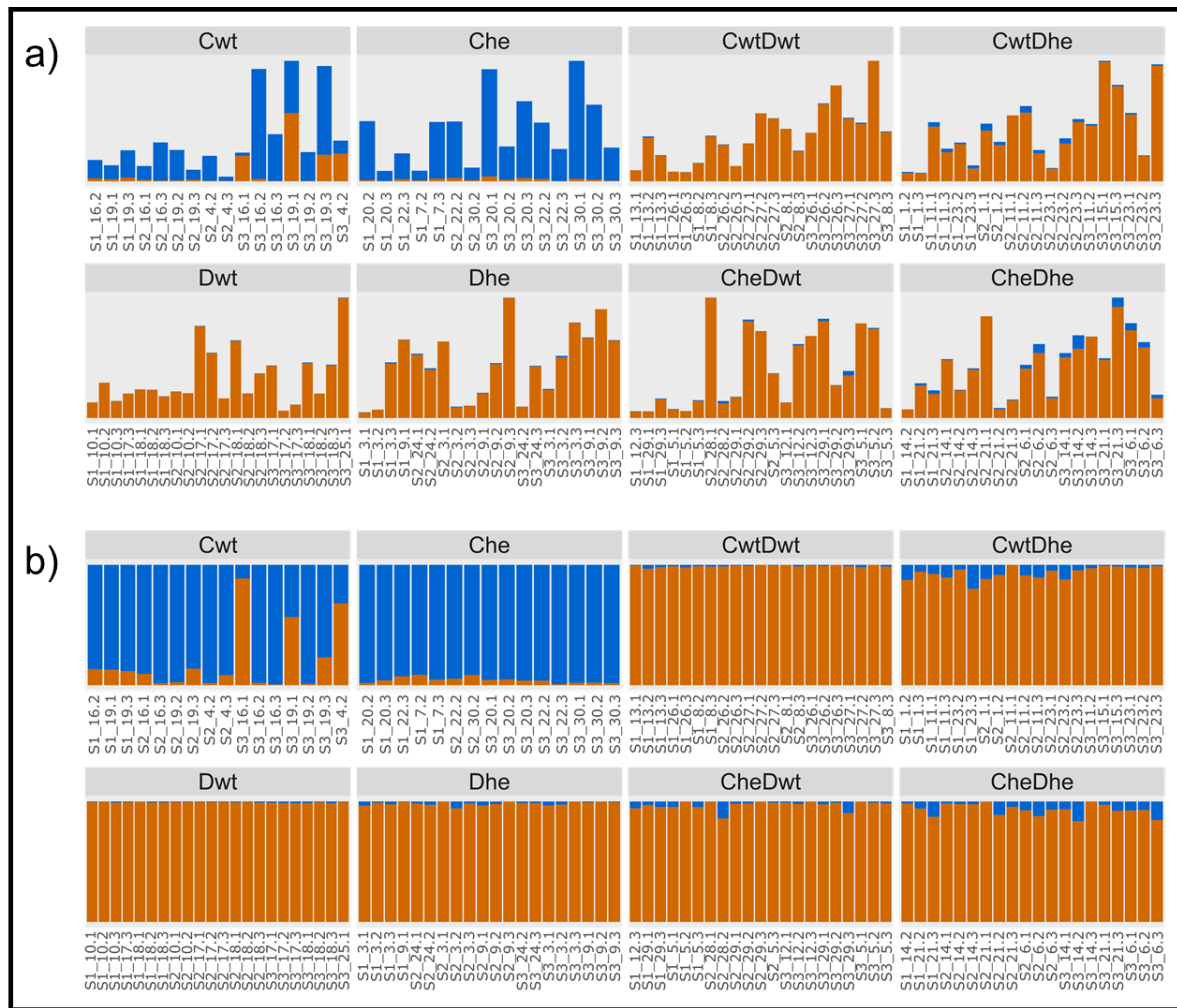


Figure S2. **a)** Absolute abundance sequencing read out of *C. proliferum* (blue) and *D. trenchii* (orange) sequences. Samples are grouped by treatment. Sample names consist of the timepoint in which the sample was taken (ex. S1), the tank number (ex. 16), and the replicate number (ex. .2). **b)** Relative abundance sequencing read out of *C. proliferum* (blue) and *D. trenchii* (orange) sequences. Sample names follow the same convention as above.

Table S1. Average survival of each treatment at each time point. The number of recruits (n) refers to the number of recruits alive at the first time point.

Treatment	Recruits, n	Survival at: Day 11	Day 18	Day 24	Day 38	Day 46	Day 54	Day 60	Day 67	Day 70
Che	159	1	0.943	0.792	0.660	0.660	0.654	0.648	0.642	0.585
Dwt	254	1	0.902	0.862	0.724	0.685	0.650	0.634	0.622	0.579
Dhe	211	1	0.829	0.796	0.640	0.602	0.602	0.602	0.602	0.569
Cwt+Dwt	268	1	0.889	0.836	0.780	0.780	0.780	0.761	0.734	0.739
Cwt+Dhe	293	1	0.891	0.863	0.775	0.536	0.522	0.515	0.515	0.488
Che+Dwt	233	1	0.768	0.622	0.391	0.391	0.391	0.391	0.391	0.361
Che+Dhe	211	1	0.853	0.825	0.758	0.720	0.555	0.555	0.550	0.530

Table S2. Average recruit area of each treatment at each time point. The number of recruits (n) ranges from the number of recruits alive in the first time point to the number of recruits alive at the last time point.

Treatment	Recruits, n	Size (mm ²) at: Day 11	Day 18	Day 24	Day 38	Day 46	Day 54	Day 60	Day 67	Day 70
Che	159 - 93	0.128	0.151	0.168	0.280	0.451	0.549	0.632	0.731	0.835
Dwt	254 - 147	0.184	0.230	0.286	0.456	0.610	0.996	1.162	1.286	1.404
Dhe	211 - 120	0.182	0.196	0.268	0.380	0.703	0.980	1.217	1.372	1.536
Cwt+Dwt	268 - 198	0.189	0.194	0.259	0.477	0.933	1.309	1.659	1.877	1.959
Cwt+Dhe	293 - 143	0.172	0.194	0.242	0.468	0.691	0.913	1.030	1.055	1.107
Che+Dwt	233 - 84	0.217	0.240	0.278	0.377	0.581	0.789	0.993	1.162	1.288
Che+Dhe	211 - 112	0.176	0.196	0.261	0.485	0.836	0.996	1.173	1.296	1.426

Supplementary Methods

Collection & Spawning

Nine colonies of *Platygyra* sp. aff. *daedalea* (most similar to *Platygyra daedalea* colonies described from the red sea; hereafter referred to as *P. daedalea*). were collected from Falcon Reef (-18.767116, 146.534248) (collection permit: G21/45348.1) prior to the November full moon 2022. They were brought to the Australian Institute of Marine Science (AIMS) and held in outdoor tanks at the National Sea Simulator (SeaSim). The colonies were isolated in 80 L aquaria on the third, fourth, and fifth nights following the full moon for approximately 30 min before and after their known spawning time (Babcock et al., 1986). An additional five colonies, collected from the same location (collection permit: G21/38062.1) and held in SeaSim over the past two years, were also isolated. Thirteen colonies spawned on the 12th of November. Four corals spawned for a second time on the 13th of November. Spawning, larval rearing, and settlement followed the well-established protocol from Pollock et al., 2017. Briefly, gametes were pooled, fertilised, and washed in 80 L aquaria. Fertilised

embryos were reared in specialised vats with a constant supply of 1 μm filtered seawater until they were deemed competent to settle.

Settlement

Thirty pre-conditioned, autoclaved concrete tiles (28cm²) were placed into separate 30-litre tanks in a controlled experimental room at the SeaSim. Each tank was kept in one of two water baths that were maintained at 28°C with non-recirculating seawater. The tanks themselves were supplied with 28°C, 1 μm filtered seawater at a rate of ~15L/h. The lights were custom-made 300w LEDs (SeaSim2G), set on a 12-hour sine curve that ramped up in the morning to deliver a maximum of 60 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ for ~6 hours before ramping back down. Sterile glass beads surrounded the tiles to prevent the larvae from settling on the sides or underneath the tiles, and netting was added to the outflow of each tank so larvae would not be lost down the drain.

One thousand competent larvae were added to each tank on the 18th of November from the 13-colony bulk culture. Settlement was assessed visually as the recruits were auto-fluorescent under a blue light. Following lower-than-expected settlement (~4% compared to the ~30% found in Diaz-Pulido et al., 2010 under similar conditions) after two days, an additional 350 larvae were added to each tank from the same 13-colony bulk culture produced on the 20th of November. A final round of 100 larvae (from the 4-colony bulk culture) was added to each tank on the 23rd of November. On the 25th of November, the number of settled coral recruits on each tile was counted and recorded, again using a blue light torch. Each tile was then assigned to an experimental group and effort made to balance the number of recruits in each treatment (212 ± 4 recruits per treatment with a range of 18-150 corals per tile).

Experimental Treatments & Inoculation

Prior to inoculation the genus of each culture was confirmed using a restriction fragment length polymorphism (RFLP) protocol developed in house (L.M. Peplow, unpubl.). One ml samples of pure culture were pelleted by centrifugation at max speed for 5 min. The cell pellets were washed in filtered seawater and resuspended in 100 μL TE buffer, then heated to 95°C for 10 min to lyse the cells. After heating, the cell suspension was briefly centrifuged and 1 μL of the supernatant was used as template for PCR amplification of the ITS2 region with ITS_DINO_F (GTGAATTGCAGAACTCC) and L_O_R (GCTATCCTGAGRGAAACTTCG) (Pochon et al., 2001) primers in a standard 15 μL reaction using Qiagen's multiplex mastermix. Next, the PCR product was subjected to a double digestion with HindIII and PstI restriction enzymes and incubated at 37°C for 2 hours. Each sample was then loaded onto a 2.5% agarose in 0.5xTBE gel next to pre-identified (via ITS2 sequencing) culture samples of *Cladocopium* and *Durusdinium* standards. Standards for *Symbiodinium*, *Breviolum*, and *Gerakladium* were also run to ensure no contamination occurred in culture prior to inoculation. All pure culture samples displayed banding patterns that were consistent with expectations for each genus, and none showed any signs of contamination with other genera (Figure S1).

Experimental tanks were inoculated on the 26th of November. The volume of water in each tank was dropped to 5 L, and their assigned symbionts were added at a density of 10,000 cells/ml. The culture volume needed for each treatment was calculated by performing cell

counts on each culture before inoculation (see Supp. Methods). Cell counts were performed on a flow cytometer (BD Accuri™ C6 Plus Flow Cytometer, BD Biosciences), following the protocol described in (Scharfenstein et al., 2023). Briefly, each culture flask was resuspended, and a 970 μL sample was taken and preserved with 20 μL of 25% glutaraldehyde and 10 μL of 1% Pluronic F-68. Each sample was sonicated and vortexed before 50 μL was processed at a speed of 35 $\mu\text{L}/\text{min}$ and excited at 488 nm. The resulting events were gated so that only events emitting at 675 ± 25 nm were counted, ensuring our numbers included only Symbiodiniaceae. Each sample was processed twice and averaged to obtain final cell counts.

Mock Communities

The observed and expected proportion of *C. proliferum* sequences in the mock communities showed no statistically significant differences ($p\text{-value} < 0.05$) (Figure 2a), indicating that there is no significant disparity in ITS2 copy number between the two species (*C. proliferum* and *D. trenchii*) used in the inoculations.

Single Strain Inoculations

Analysis of the ITS2 sequences from the single inoculation treatments at time point 1 indicated that there was some background contamination ($2.7\% \pm 0.9$ SE, $n = 18$) present in $\sim 85\%$ of samples (Figure S2). Further analysis of the number of contaminating reads in each single inoculation sample and their associated negatives suggested that this contamination likely arose during sequencing preparation rather than being genuine reads from their respective samples (Kircher et al., 2012; Kotrova et al., 2017). The largest supporting evidence for this explanation is the discovery of a unique DIV (defining intragenomic variant) that is typically only present in samples taken from the field. This DIV was not found in any of the symbionts offered to our coral recruits, but it was present in other coral samples that were sequenced in the same run. This indicates that there was some contamination during library prep, as manual handling of untagged PCR products is prone to sample-to-sample contamination (Hess et al., 2020).

Unfortunately, one treatment, **Cwt**, at timepoint 3 showed a very high number of reads from *D. trenchii* ($33.3\% \pm 13.9$ SE, $n = 7$). When compared to the number of *D. trenchii* reads in the previous timepoint ($1.01\% \pm 0.68$ SE, $n = 6$), as well as the contaminating reads in all other single inoculation treatments ($0.7\% \pm 0.2$ SE, $n = 23$), it can be concluded that this contamination is likely genuine. Therefore, **Cwt** was excluded from all further analyses.

Supplementary References

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